Project 1 - Sparsity of E-Commerce Data

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General Procedure

- **Organization (for each phase):**
  - Individually work on easy bullet points
  - Work together on the more difficult bullet points
- **Implementation:**
  - Used Java with JDBC for database access local PostgreSQL server.
  - Generated test data programmatically
  - Implemented horizontal and vertical representations
  - Implemented transformations between both representations
  - Benchmarked different query strategies

Schedule

- In general followed the deadline schedule
  - Worked well for Phase 1
  - Worked well for Phase 2
  - We underestimated the effort needed for Phase 3, leading to some stress near the end

What went well, what not?

- **Went Well:**
  - Phase 1; setup, toy and generate() were mostly straightforward
  - Benchmarking was somewhat time-consuming, but uncomplicated
- **Did not go well:**
  - We were not overly familiar with non-natural joins in SQL
  - Initial correctness of SQL procedure approach (`q_i(oid)`, `q_ii(attribute, value)`.
  - Choosing the right plot(s) to visualize the huge benchmark results

Benchmark

- PostgreSQL 18.3, Java 25.0.2 (JDBC 42.0.6)
- Client & Server on same machine
- **Machine:**
  - **CPU:** Ryzen 5 5600H 6-Core 12-Threads 3.3 Ghz
  - **RAM:** 16GB, 3200 MHz DDR4
  - **OS:** Fedora Linux 43
- **Parameters:**
  - $\#\text{tuples} \in \{2000, 4000, 8000\}$
  - $\#\text{attributes} \in \{5, 10, 15\}$
  - $\text{sparsity} \in \{0.5, 0.75, 0.9375\}$
- Each combination repeatedly executes the following two queries in 10s each:

```
SELECT * FROM H_VIEW WHERE oid = <value>;
SELECT oid FROM H_VIEW WHERE <attribute> = <value>;
```

Conclusion

- Choosing between horizontal and vertical is highly query-specific:
  - Query 1 performs very well for vertical representations
  - Query 2 performs badly for vertical representations, unless the sparsity is high and #attributes is low
- Vertical representations perform better, the less attributes they have and the higher the sparsity is
- High sparsity relations are not necessarily more space-efficient in vertical representation
- Vertical may be worth it for the its other benefits (Schema Evolution)